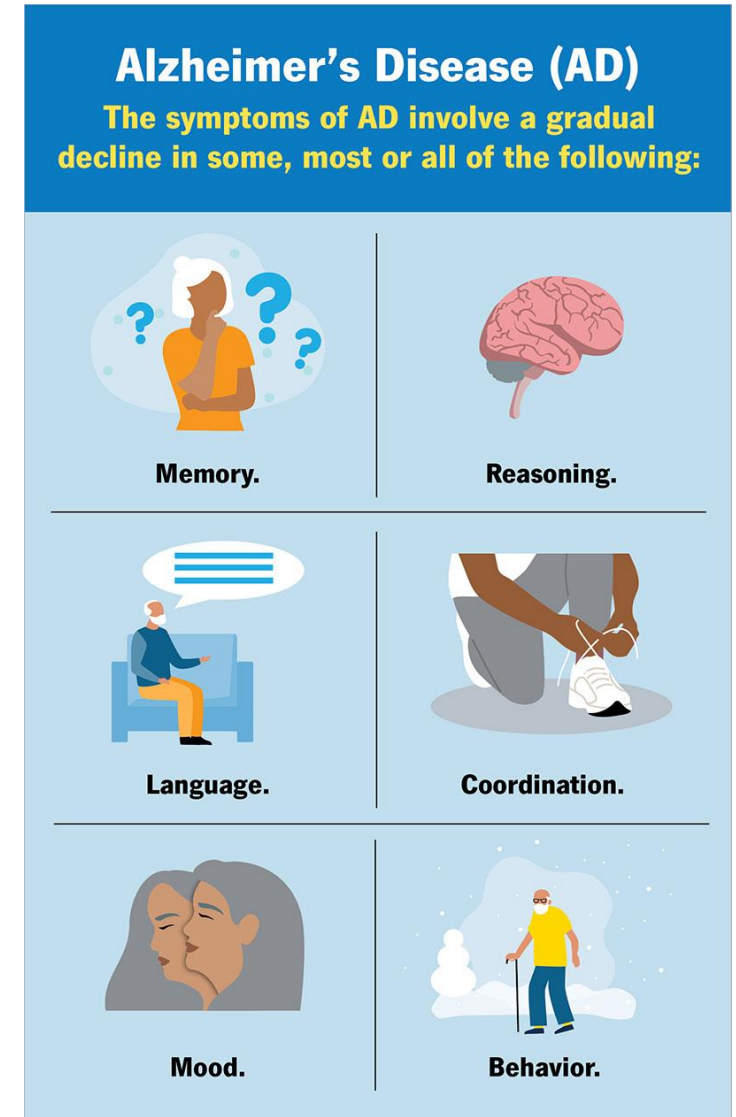


# Deep Learning-Based Animal Behavior Analysis: Insights from Mouse Models of Alzheimer's Disease

Papers : <https://arxiv.org/abs/2602.09518>  
<https://arxiv.org/pdf/2508.05138>  
Project page : <https://github.com/franktpmvu/Universal-Action-Space.git>

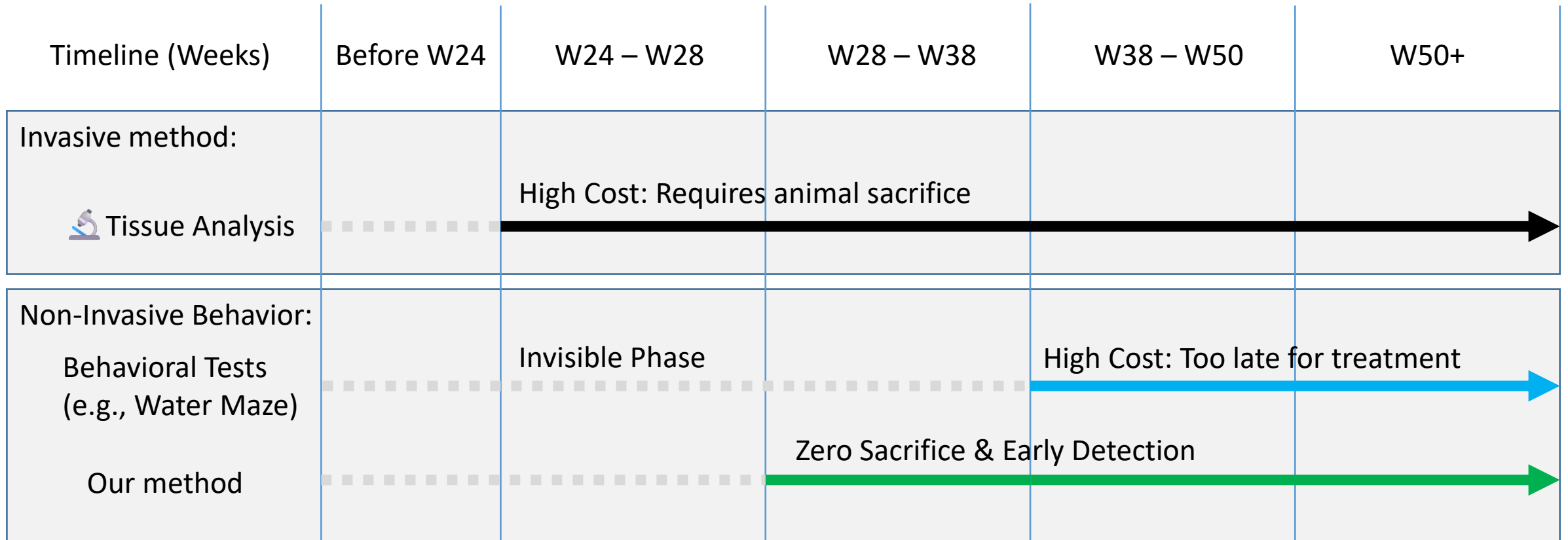
# The Stages of Alzheimer's Disease (AD) in Humans

- **Early stage:** short-term memory decline, word-finding difficulties, mild disorientation, anxiety/depression.
- **Middle stage:** getting lost in familiar places, difficulty speaking and understanding, increasing dependence in daily activities.
- **Late stage:** failure to recognize family members, severe language impairment, motor function decline, fully dependent on care.



# The Difficulty of Early Detection in Alzheimer's Disease in Mouse Models

- A critical time lag exists between invasive biological markers and observable behavioral symptoms.
- Subtle and gradual behavioral changes that are hard to quantify



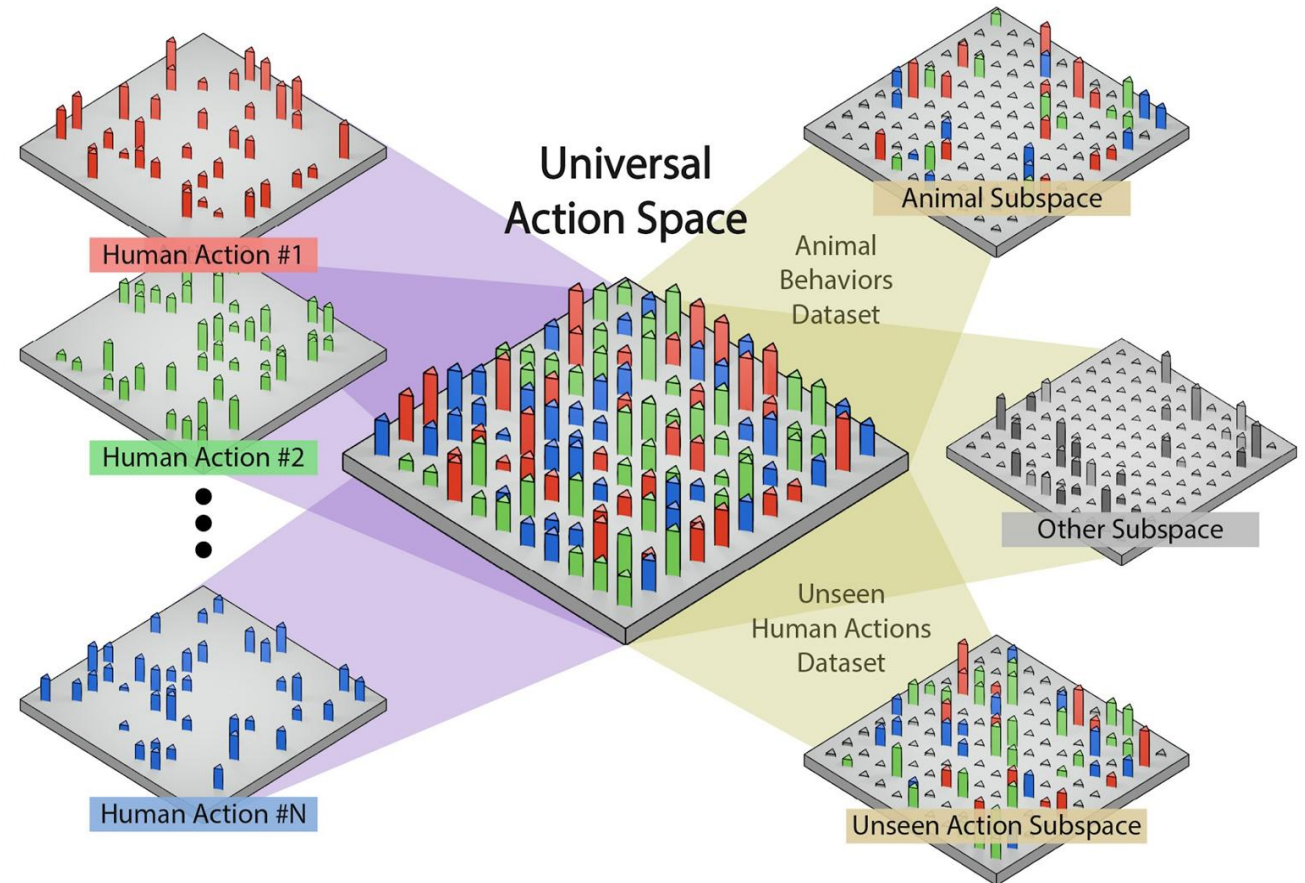
Undetectable ■ ■ ■ Detectable —

# AI-Powered Behavioral Phenotyping

- Decoding Alzheimer's symptoms within the "Invisible Phase" using periodic 5-minute multi-view snapshots.

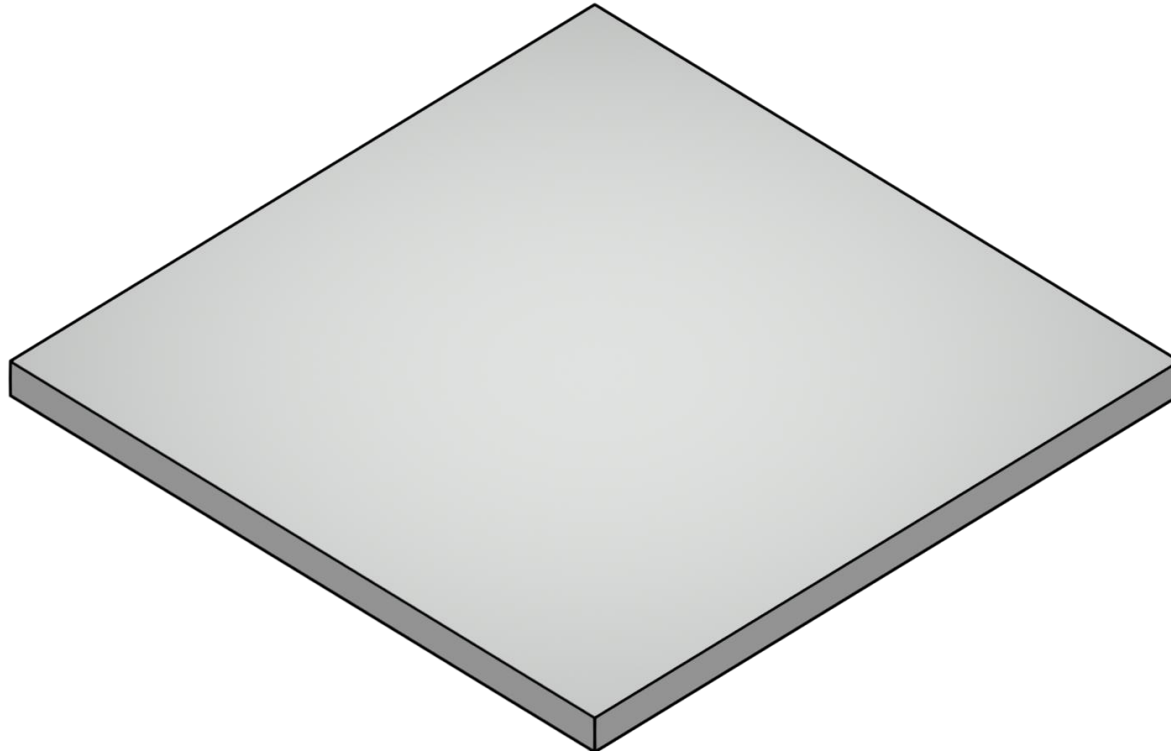


- 5-Min Bi-weekly Snapshots
- Dual-View Integration
- Dataset: 128 Standardized Videos (8 AD / 8 Control)



- Feature Extraction:** Compressing 5 minutes of motion into a "Universal Action Space."

# Construction of UAS

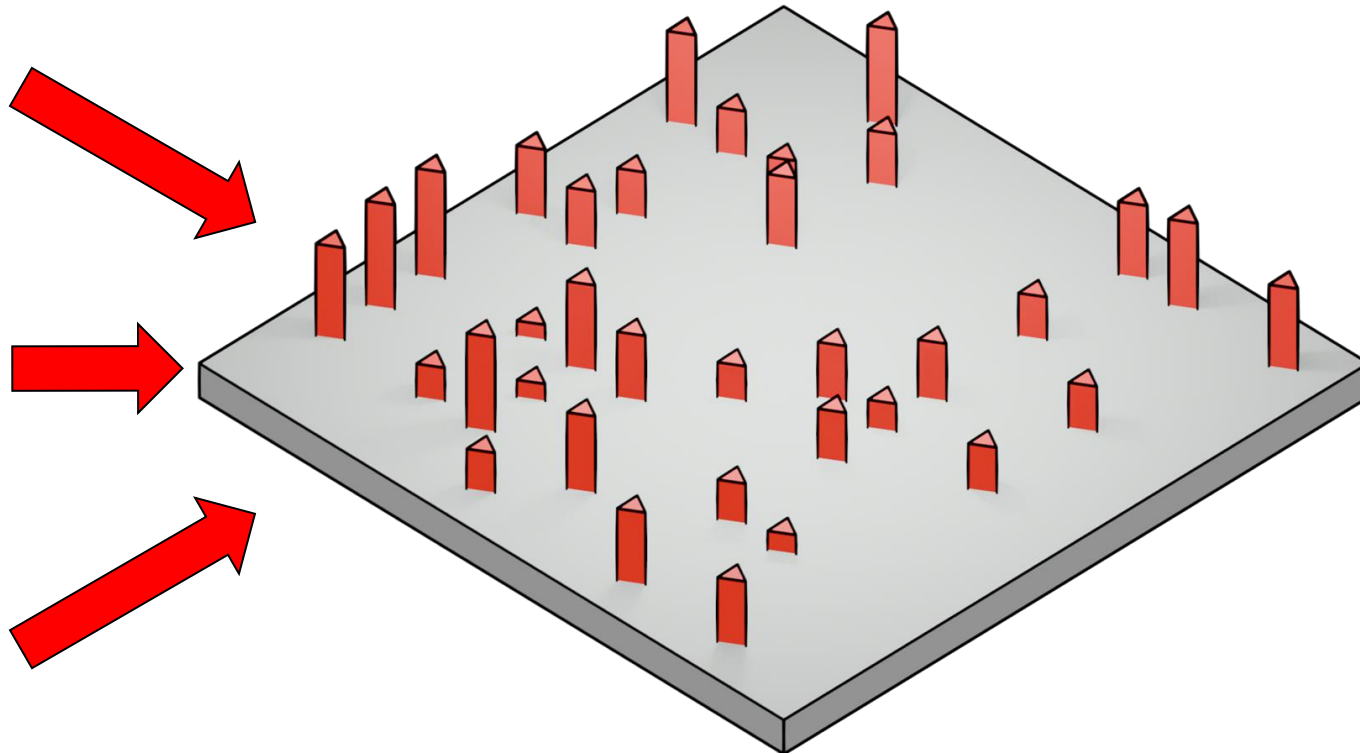


# Construction of UAS

Riding horse



⋮



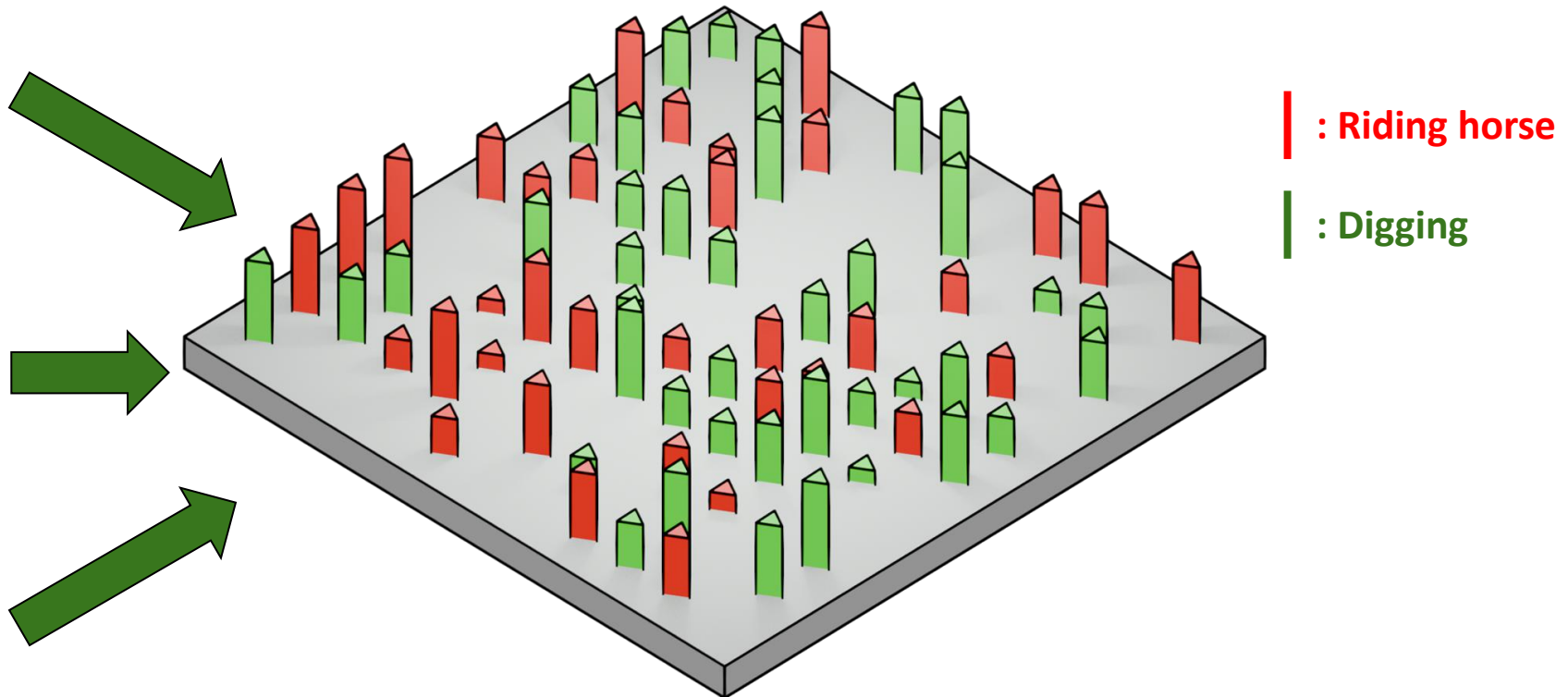
| : Riding horse

# Construction of UAS

Digging



⋮



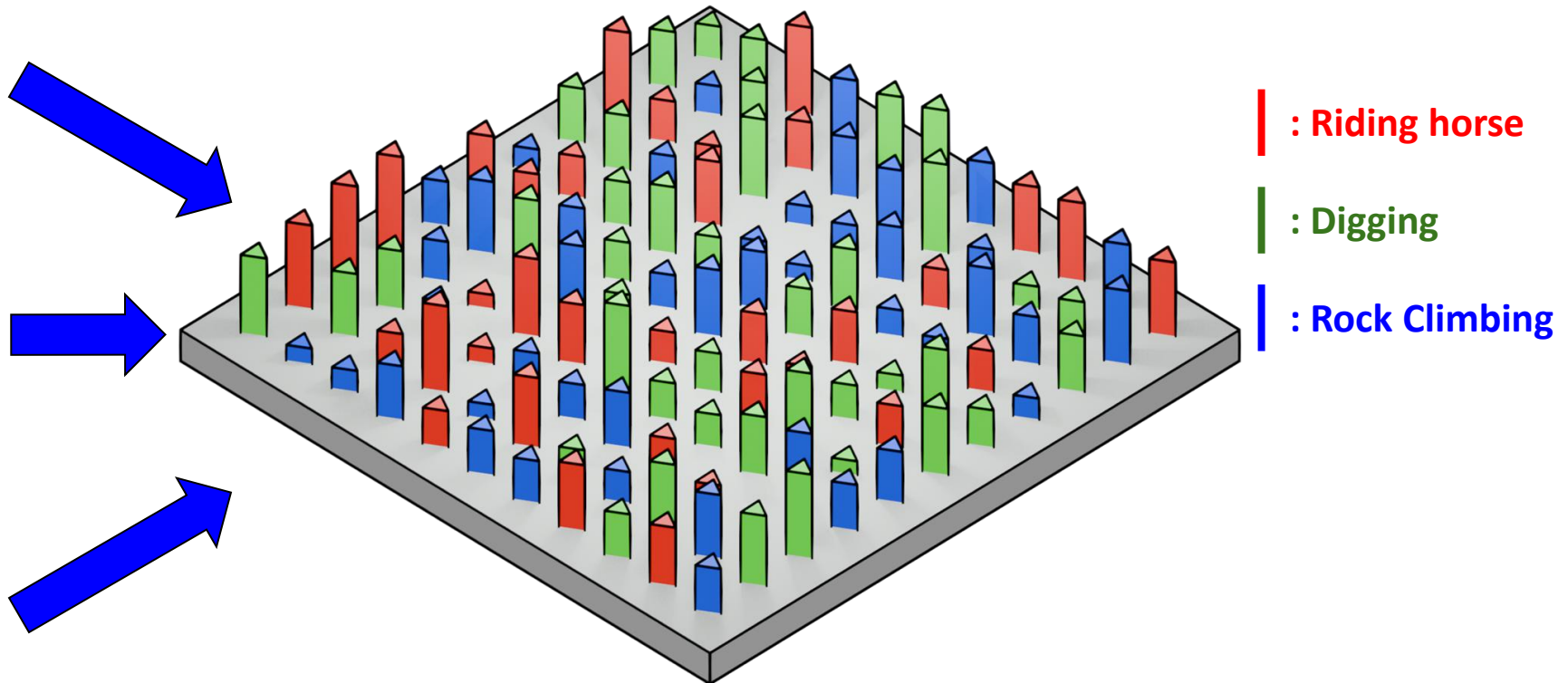


# Construction of UAS

## Rock Climbing

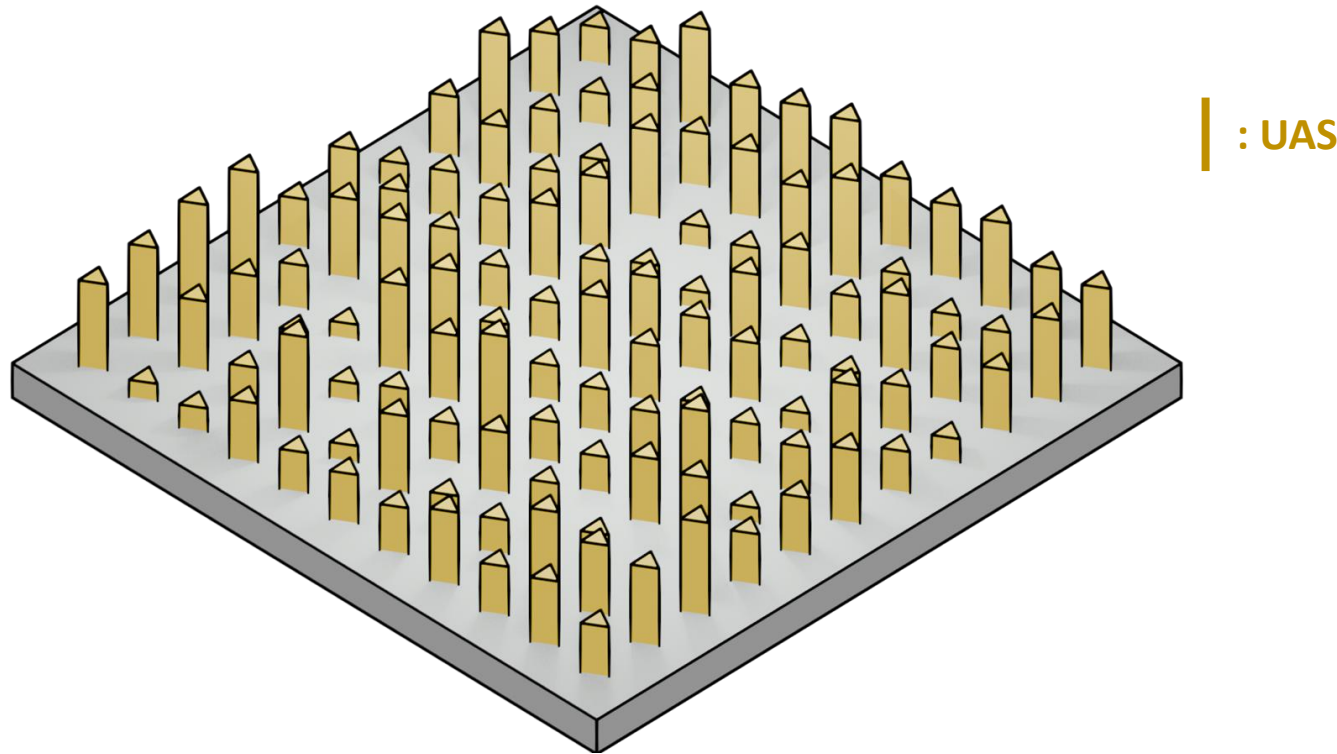


⋮





# Construction of UAS

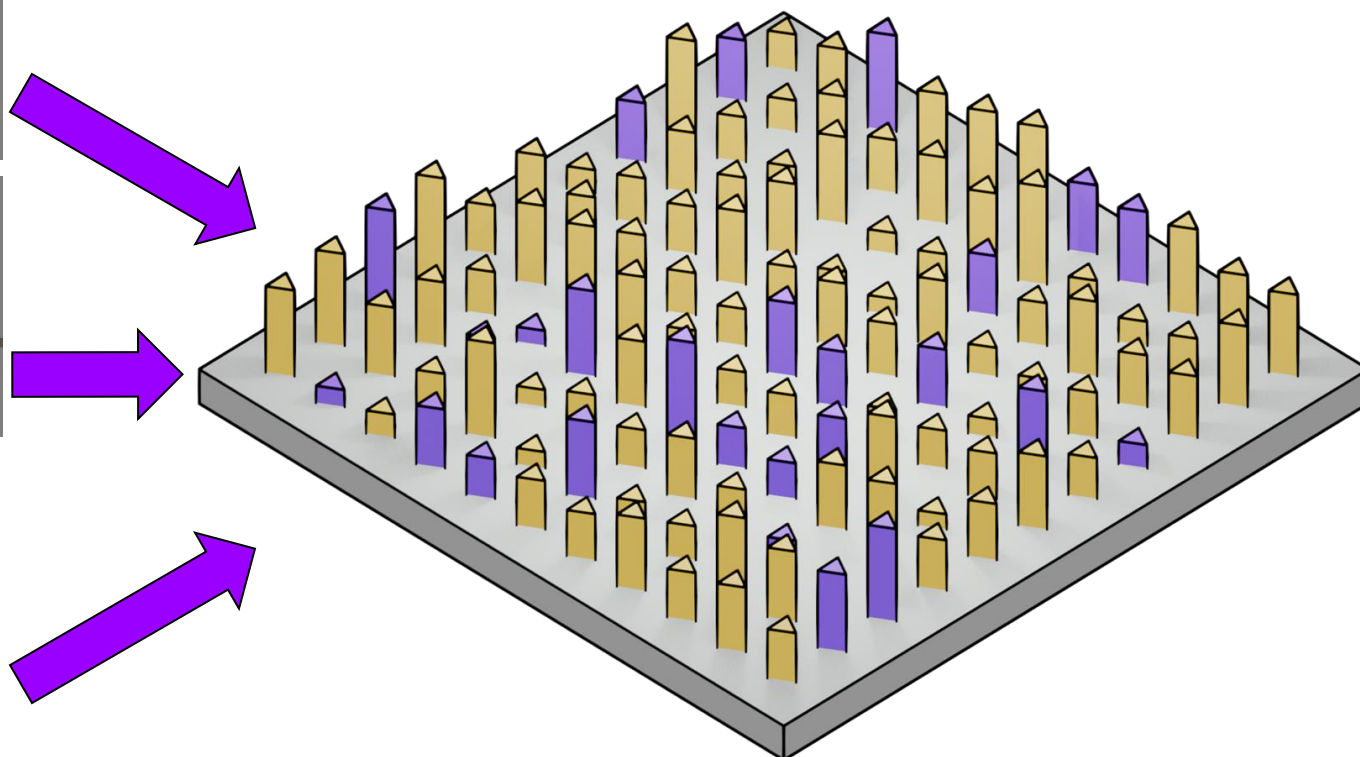
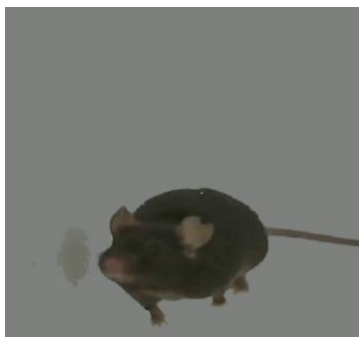


# Cross-Species Feature Mapping

Mouse behaviors



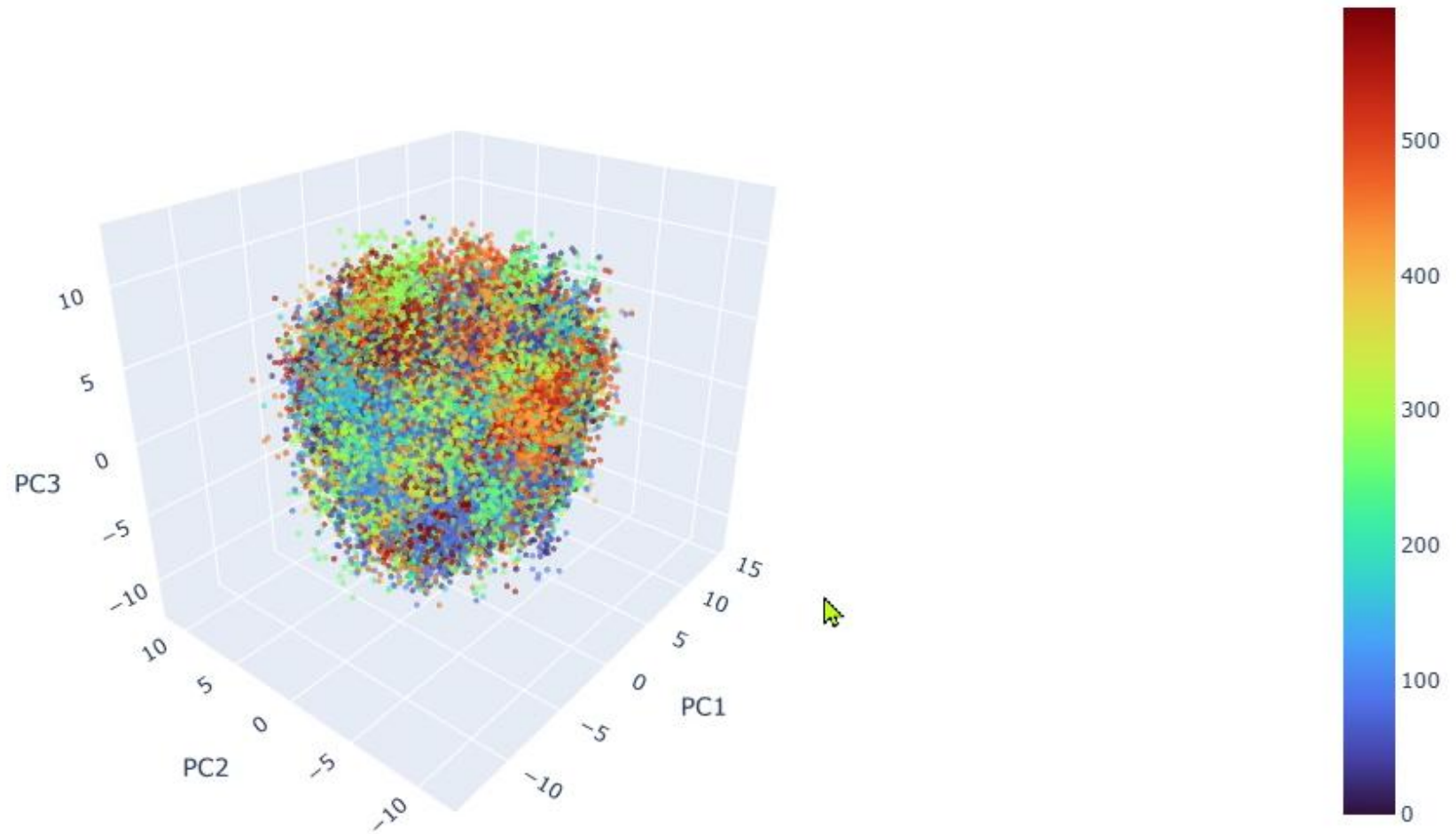
⋮



| : UAS

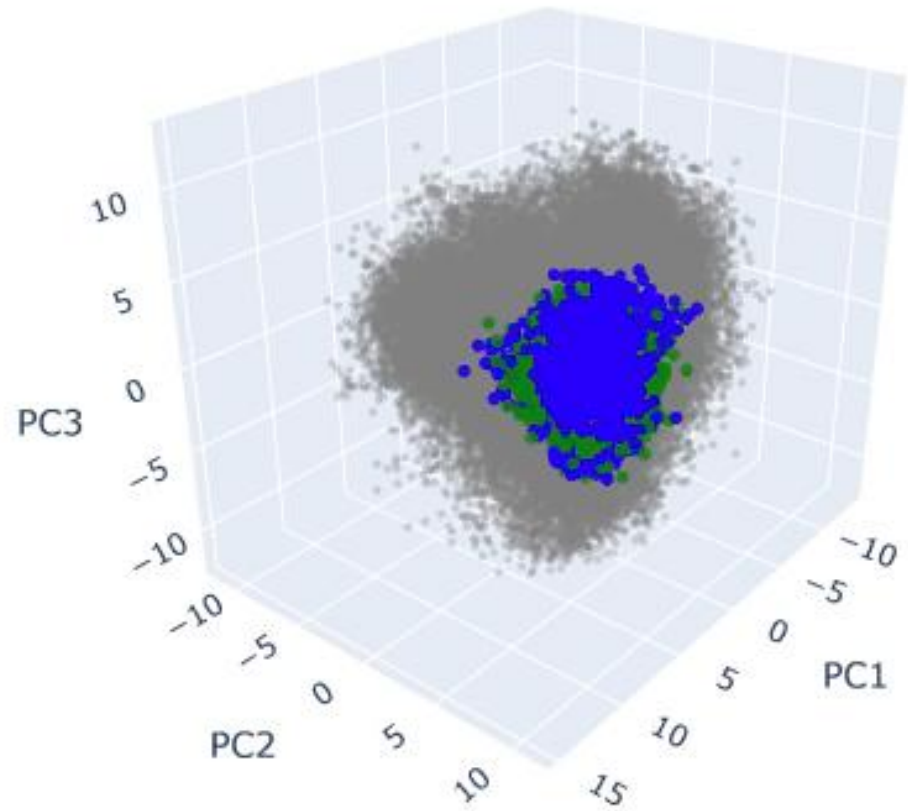
| : Domain-Specific subspace

# Visualizing the True UAS Embedding Space by PCA



Each point represents a 1.5s video clips from K600.  
Colors indicate different action classes.

# Cross-Species Alignment: Mouse Behaviors in Human UAS



Gray: Human Base (K600 Space)  
Blue: AD Mouse Data  
Green: Healthy Mouse Data

Mouse behavioral patterns are fully captured within the Human UAS.

# UAS-Powered Classifier

Unseen actions

Backbone

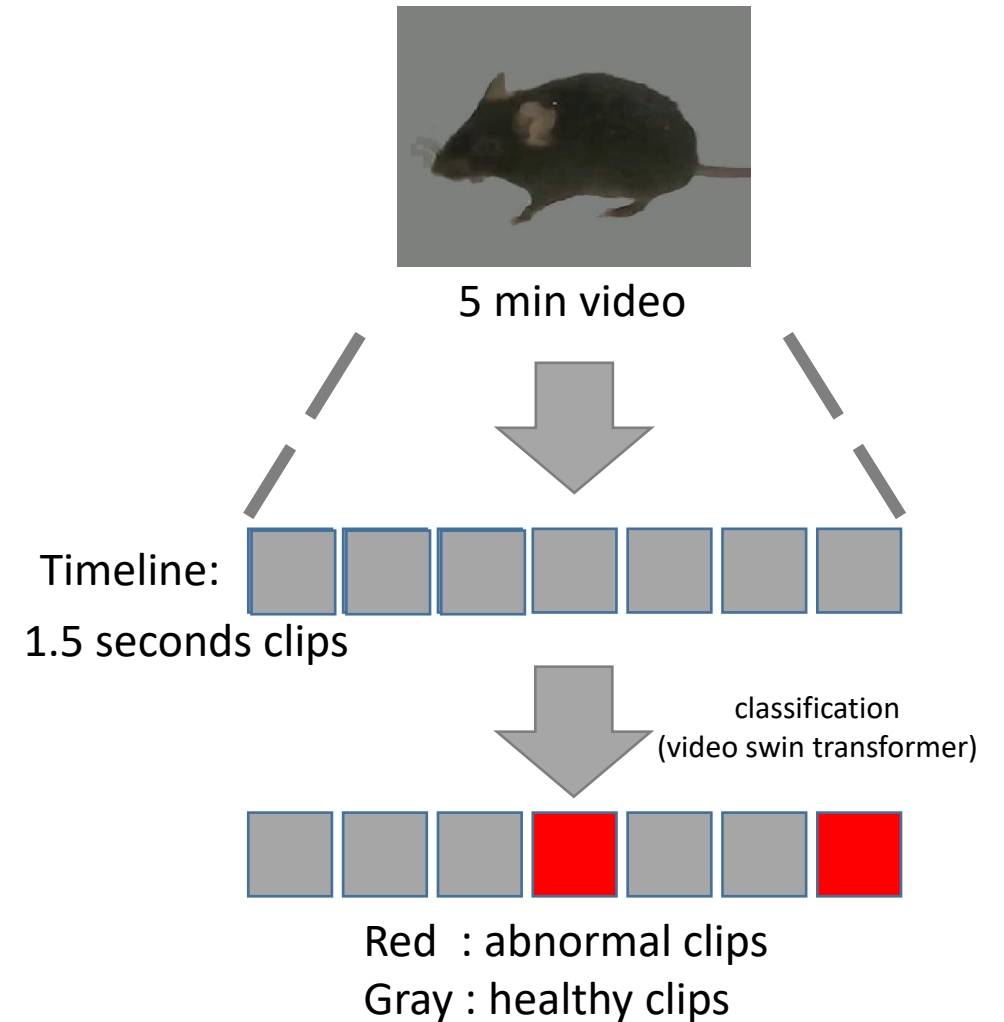
Head

Video Swin Transformer

Classifier

Action Class

Universal Action Space



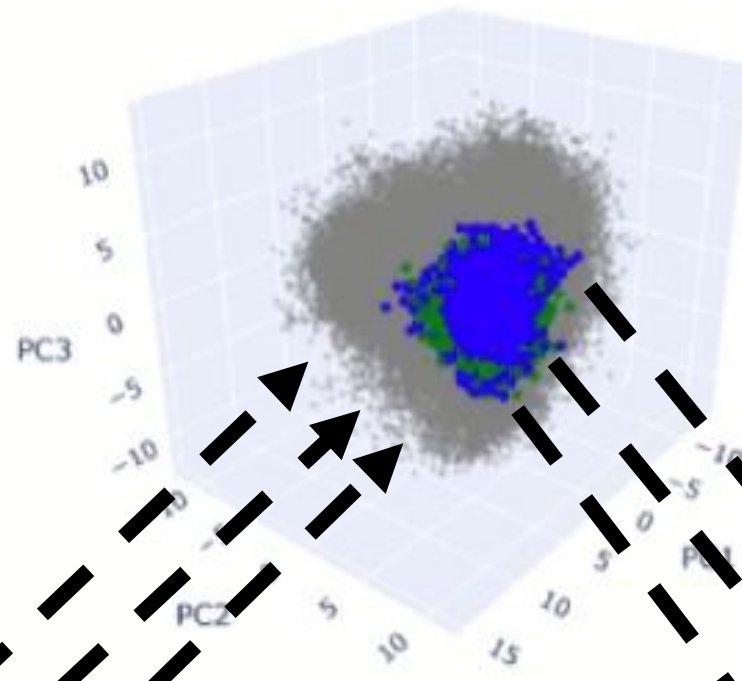
# UAS Feature Mapping & Classification



5 min mouse video



Maps clips into the UAS feature space.

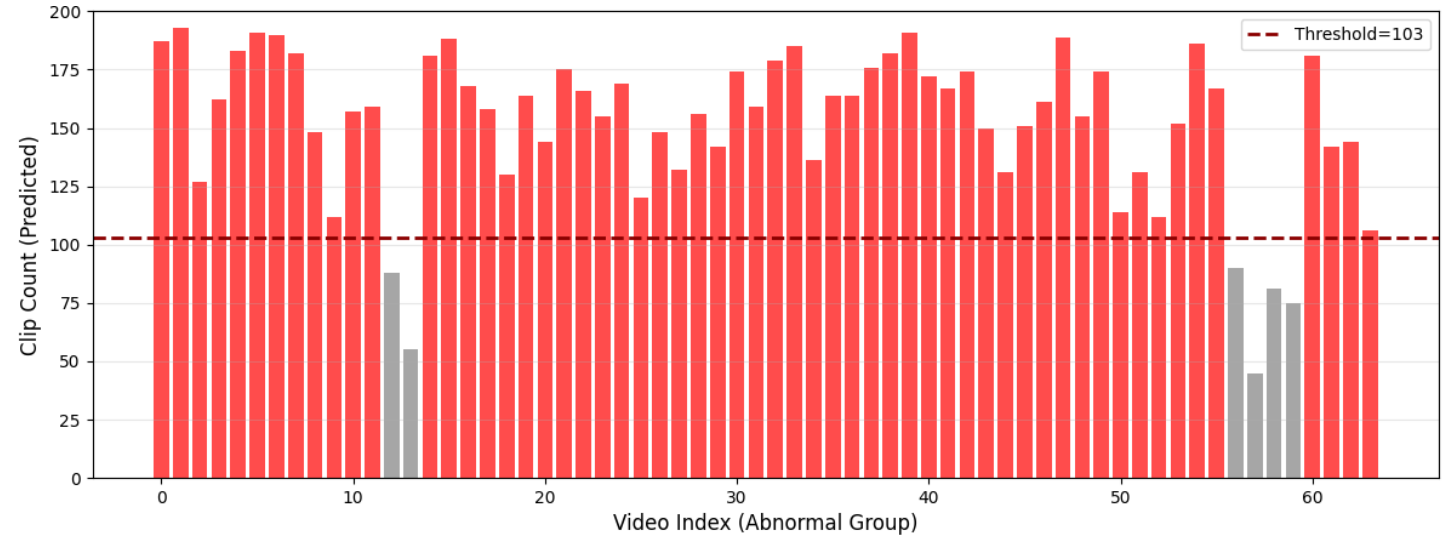


Classifies abnormal behaviors via a non-linear FC layer (latent similarity computation).

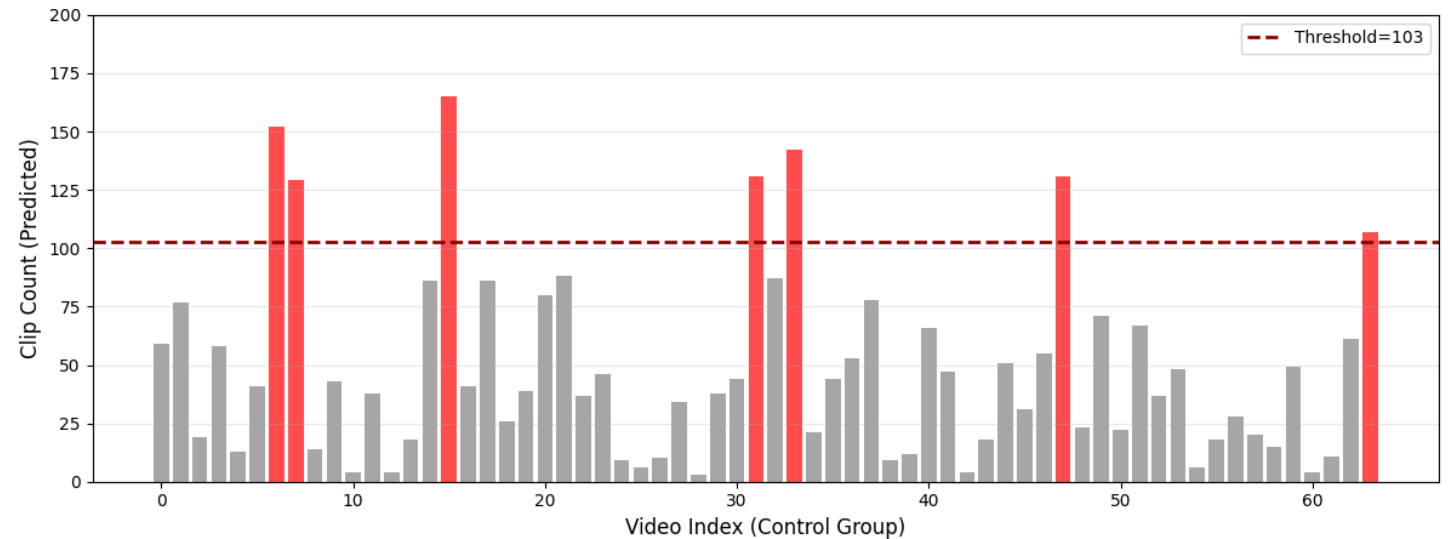


# Statistical Evidence

- **X-axis:** Individual Video Index
- **Y-axis:** Detected Abnormal Clips
- **Red Line:** Optimized Threshold ( $k=103$ )



Abnormal clips in AD videos



Abnormal clips in Healthy videos

# Results

- Threshold optimized via Youden's J statistic with Precision-Recall curve.

|      | Precision | Sensitivity (Recall) | Specificity | Top-1 accuracy | Earliest detection weeks |
|------|-----------|----------------------|-------------|----------------|--------------------------|
| Ours | 94.38     | 88.30                | 95.08       | 91.80          | 28                       |

